Exathon 2026, Track 01: Money, Risk & Regulation

Declined-Application Explainer

Every decline should come with a reason and a way forward.

A decline today ends the conversation

The customer

Gets a one-line rejection: no reason, no next step, and no idea whether reapplying is worth it.

The lender

Branches and call centres field repeat questions, complaints and escalations about decisions they cannot easily explain.

The regulator

Decisions need to be explainable and defensible, and shown to be applied fairly.

From a score to a letter in four steps

1 Score

An interpretable logistic regression rates the application on six non-sensitive factors.

2 Reasons

Each factor's pull against an average applicant. The biggest negatives become the stated reasons.

3 Path to approval

The smallest realistic single change that would flip the decision, such as lowering card usage.

4 Letter and audit

A plain-language letter, plus the structured facts behind it for any reviewer.

One applicant, end to end

Declined

Approval score 43%; 50% is needed.

Credit score 690
Card usage 95%
Obligations 45% of income
Loan 1.5x annual income

Why

- You use 95% of your available revolving credit.
- Your credit score of 690 is below what we need.
- Obligations would take 45% of your income.

What could change it

- Card usage from 95% to about 87%.
- Obligations from 45% to about 42%.
- Credit score from 690 to about 700.

Any one of these may be enough on its own.

Built to be defended, not just deployed

No sensitive attributes

Age, gender, religion, caste and pincode are never model inputs. A test enforces it.

AI only rephrases

Claude rewrites facts already computed. It cannot add reasons, numbers or promises, and the template takes over on any error.

Human review, always

Every decline letter states the customer's right to ask a person to review the decision.

Honest limits

If no realistic single change would help, the letter says so instead of inventing advice.

Simple stack, fully testable

Layer	Choice
Interface	Streamlit app hosted on Replit
Model	Logistic regression on standardised features
Data	Synthetic applications in INR; no real customer data
Wording	Template by default, optional Claude rewrite
Quality	Unit tests for accuracy, explanations and fairness

Live demo: declined-application-explainer--nirmal-04.replit.app

What comes next

Real-world data

Train on bureau and repayment history, with proper validation and bias testing.

Every customer

Letters in Tamil, Hindi and other regional languages, by SMS, email and WhatsApp.

Lender workbench

Reason codes mapped to credit policy, a human-review queue and decision logs for audits.

Every decline: a reason, a path, a record.